Perhaps most importantly, in MAB, we face the same exploration–exploitation tradeoff that we discussed in Section 4.1.2. In particular, each time we decide to pull the arm of a machine, we need to choose between the machine that has been giving us a good success rate so far (exploitation) and those whose reward probability we don't know too much about (exploration).

The problem we face is a tradeoff because by exploring, we run the risk of wasting our pulls on machines that have low reward probability, but excessive exploitation means that we might completely miss out on a machine that has a higher reward rate than what we currently observe.

Figure 5.3 shows an example of this, where:

1. The first machine offers the highest empirical success rate so far at 70 percent.
2. We have more data from the second machine, so we have the least uncertainty about its reward rate, which is around 50 percent.
3. Although the empirical success rate of the third machine is the lowest (0 percent), we may need to try pulling its arm more in order to be more certain of its reward rate.

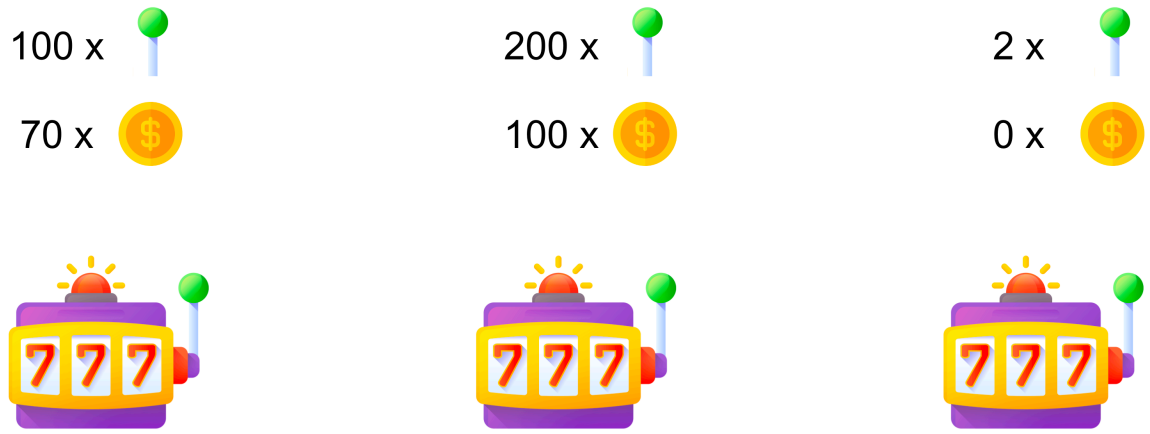


Figure 5.3 An example of a MAB data set showing the exploration/exploitation dilemma. A MAB policy must choose between a machine with a consistently high success rate and one whose reward rate is uncertain.

It is the job of a MAB policy, similar to that of a BayesOpt policy, to look at the data on past rewards and decides which arm we should pull next, while balancing this tradeoff between exploration and exploitation.

The MAB problem models a wide range of applications that you might see in the real world:

- In product recommendation, an engine needs to pick from a number of products in store to suggest to a user. Each product can be seen as the arm of a slot machine, and pulling the arm means that the engine picks that product to present to the user. If the user clicks on the advertisement for the product, we may see it as receiving a reward, as the user's click is something we want to achieve.

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